

Mostafa Elsayed

Mobile Engineer — Flutter & Dart | 3+ Years

Real-Time Systems · OCPP / EV Charging Infrastructure

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[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

Mobile engineer with 3+ years and five apps live on the App Store and Play Store. Built the driver-facing application for an EV charging network on OCPP 1.6J — the same open protocol running most operational charging infrastructure across Europe — handling the full transaction lifecycle from StartTransaction to StopTransaction, live MeterValues, connector state transitions (Available → Preparing → Charging → Finishing → Faulted), Heartbeat-based liveness detection, and session-state recovery on reconnect. Work in Clean Architecture with Bloc/Cubit, unit-test critical paths with bloc_test and Mockito, and ship through automated CI/CD (Fastlane, Codemagic, GitHub Actions). Also ship production web with Next.js 15.

EXPERIENCE

Flutter Engineer (Mid-Level)

Jan 2025 – Present

Easy Touch IT Solutions (Al Manarah Holding Group), Abu Dhabi, UAE — EV charging infrastructure & interactive display products

- Built and shipped **Easy Touch (EV Charging)**, the driver-facing app for an EV charging network, live on the App Store and Play Store. Implemented the full **OCPP 1.6J** charging flow over WebSocket: StartTransaction/StopTransaction lifecycle, live MeterValues streaming, connector state transitions (Available → Preparing → Charging → Finishing → Faulted), and Heartbeat-based liveness detection.
- Designed the **session billing logic** on top of live meter readings — drivers start a session by target energy (kWh), by target spend, or open-ended against their in-app wallet balance, with automatic conversion between energy and cost and session termination at the correct threshold.
- Built **session-state recovery for unreliable connectivity** — on WebSocket drop mid-charge, the app reconnects and reconciles live session state rather than losing or duplicating an active transaction.
- Unit-tested Cubit state transitions and OCPP message handlers with **bloc_test** and **Mockito**, covering the charging state machine and its failure paths.
- Built **Easy Touch (Interactive Display)**, the whiteboard application for the company's **75"/86" interactive screen product** — freehand drawing at **60 FPS on embedded Android** using CustomPainter and the Canvas API, Riverpod for state, and Hive persisting vector paths for near-instant save/load. Integrated with an AI backend that recognises a sketch and returns the closest clean geometric shape.
- Rebuilt and relaunched the **Quarak QR** app, replacing an outdated Play Store listing with a modern production release.
- Shipped five production websites with **Next.js 15**, including a bilingual AR/EN build with full RTL — handled SEO, Google Search Console, and deployment. ([diamondsword.ae](#) · [futuregulf.ae](#) · [dyfmotors.ae](#) · [commercialoxforduae.ae](#) · [westernbeach.ae](#))

Mobile Application Developer

Feb 2023 – Dec 2024

Esmos Tech, Cairo, Egypt — software outsourcing; white-label delivery for international clients

- Delivered production Flutter apps across e-commerce, healthcare and rental for international clients under white-label agreements, using Clean Architecture for modularity and testability.
- Built offline-first data layers with **Hive** for NoSQL caching and **SQLite** for relational data, keeping listings and search responsive on poor connectivity and syncing on reconnect.
- Built mapping and location features with Google Maps — custom markers, polylines, and location-driven screens.
- Handled end-to-end payment integration and secure API communication, with multi-currency and localised payment support.
- Built the Flutter client for an AI-driven fitness and nutrition product that generates per-user nutrition plans from workout data, integrating with a Node.js/MongoDB backend built by a separate team.

SELECTED PROJECTS

Easy Touch — EV Charging — Flutter, Cubit, Dio, WebSocket, OCPP 1.6J, Google Maps, bloc_test, Mockito [App Store](#) · [Play Store](#)

Real-time charger map and live session management for an Abu Dhabi charging network, with wallet-based and threshold-based billing over live MeterValues, and state recovery across connection drops.

Smart Greenhouse — Real-Time Device Control — Flutter, GetX, Socket.IO, MVC · 2023

Mobile control system for an automated greenhouse: live sensor telemetry (temperature, humidity, ultrasonic distance) streamed over Socket.IO, with lighting and climate actuators running on automatic rules or switched to manual override from the app. Built the full real-time control layer with low-latency command dispatch.

Easy Touch — Interactive Whiteboard — Flutter, Riverpod, Hive, CustomPainter, embedded Android

Whiteboard app for a 75"/86" interactive display product — 60 FPS freehand drawing on embedded hardware with persistent vector storage, plus AI shape recognition that snaps a sketch to the closest clean geometric shape.

GitMind — Flutter, Cubit, get_it, Firebase Auth, SQLite, Gemini AI, GitHub Actions [GitHub](#) · [Open source](#)

GitHub repository explorer with AI summaries and a context-aware chat assistant. Clean Architecture with unit-tested Cubits across cooldown and rate-limit states; cut summary prompt tokens by ~50%, handled 429 rate limits with countdown and auto-retry, and set up a GitHub Actions pipeline distributing dev builds via Firebase App Distribution.

Innvii Rent — Flutter, GetX, GetStorage, MVC [App Store](#) · [Play Store](#)

Car rental app with an offline-first listing layer — vehicle catalogue and pricing cached locally so browsing and search stay responsive on poor connectivity, syncing on reconnect.

Quarak QR — Flutter, Dart [Play Store](#)

Full rebuild and relaunch, replacing an outdated listing with a modern production release.

AI-Driven Fitness & Nutrition — Flutter, Cubit, Dio, MVVM · Client project

Flutter client generating per-user nutrition plans from workout data, integrated with a Node.js/MongoDB backend built by a separate team.

TECHNICAL SKILLS

Mobile: Flutter, Dart — mobile and embedded Android; App Store & Play Store publishing

Architecture & State: Clean Architecture, MVVM, MVC; Bloc & Cubit, Riverpod, GetX

Real-Time & APIs: OCPP 1.6J, WebSocket, Socket.IO, REST, Dio, Retrofit

Testing: flutter_test, bloc_test, Mockito, TDD, unit testing

CI/CD: Fastlane, Codemagic, GitHub Actions, Firebase App Distribution

Storage: Hive, SQLite, GetStorage, Shared Preferences, Firebase

Web & Backend: Next.js 15, React, Tailwind CSS, Node.js, MongoDB

Tools: Git, GitHub, Figma, Android Studio, Xcode; Claude Code, Cursor

EDUCATION

B.Sc. Computer Science

2019 – 2023

Misr Higher Institute for Commerce and Computers(Computer Science) , Mansoura, Egypt

Currently learning Node.js and backend fundamentals; German language (A1) in progress.